

## **Applied Machine Learning Final Project Report**

Your final report should follow the format of a standard scientific report and be submitted in either Word or PDF format. Additionally, a supporting Jupyter notebook must be included with your submission. The code in your notebook should be well-formatted and include markdown blocks describing each section of code. Please do not submit the data directly; instead, provide links where the data can be retrieved, or, if the data cannot be readily shared, include a description of where and how the data can be accessed. Aim to keep your report to a maximum of 8 pages, plus a citations page. The report should include the following sections:

### **Title**

Choose a concise, descriptive title for your project, giving an idea of its focus. List all team members on the title page.

### **Introduction**

This section introduces your project, outlining its purpose and goals. Clearly state what you are trying to achieve and identify your primary stakeholders. Additionally, explain the needs of these stakeholders that your project seeks to address and discuss the significance of these needs. Summarize your key results briefly in this section. Keep this section concise, limiting it to a few paragraphs.

### **Literature Review**

In this section, provide more detailed context about your problem. You may reuse relevant content from your proposal if applicable. Clearly establish the needs of your stakeholders and explain why you chose the methods you did, referencing prior work and the specific nature of your problem. If no one

has directly worked on your problem (an unlikely scenario), mention this and instead discuss related problems where similar methods have been applied.

## **Data**

This section will provide an overview of your dataset, including its source and a rationale for why it can be considered reliable and suitable for your project. Discuss key aspects of the data, such as the number of rows and columns, feature types, and class balance (if applicable). You may also highlight relevant details or challenges related to the dataset.

If beneficial, include 1-3 images to enhance understanding, such as visualizations of data distributions or examples of specific instances in the dataset. These images are optional but can be helpful if they illustrate challenges or insights effectively. Additional exploratory data analysis (EDA) can be included in your Jupyter notebook for further reference.

## **Methods**

Describe your methods in this section. Start by explaining any preprocessing or data transformations that were necessary (be concise here). Then, provide a detailed account of the methods you used and how they were applied. If you tried multiple methods that were unsuccessful, explain what you did and why those methods did not work. For any methods that are novel or were not discussed in class, include a clear explanation to ensure the reader understands the approach and gains confidence that you applied the methods correctly. Do *not* include results in this section; it is intended to provide an overview of the steps taken. If you attempted approaches that were not successful, explain them here, but make sure to clearly identify your final approach.

## **Results**

Share the outcomes of your work in this section. Be sure to include appropriate metrics (note: accuracy score alone is usually insufficient) and explain how these metrics were calculated—typically through n-fold cross-validation. If you analyzed feature importance or clustered your data, present those results here. This section should also include visualizations and/or tables to illustrate your findings; however, include only the results necessary to support your claims. Any additional results can be documented in your Jupyter notebook, and you can indicate where those results can be found. If you developed a prototype, include a screenshot to demonstrate it. If the prototype is accessible via a local server or the web, provide instructions on how to run the code in an appendix, a footnote, or your Jupyter notebook, but do *not* include those instructions in this section.

## **Discussion**

Use this section to reflect on your results and evaluate the extent to which you achieved your goals. Be honest in your assessment—if you believe you did not fully meet your objectives, simply state that and explain why. Importantly, critically analyze your stakeholder needs and describe how your work addresses those needs.

## **Limitations**

Please explain the limitations of your work—how it could be improved and what you would have liked to accomplish but were unable to. If you addressed some but not all of your envisioned stakeholder needs, specify which needs were unmet. Discuss ways to make your work more robust and consider whether there are any reasons to question the quality of your results or performance scores. Be critical of your own work to preempt any concerns or critiques.

**Future work**

This should be self-explanatory: What are your next steps?  
Where would you like to go from here?

**Citations**

Citations page using MLA format.